Name: ______________ **Reg No:** ______________ **Quiz #5**

1. **One-hot encoding of a word with vocabulary size $|V| = 50,000$ has:**
   a. 50,000 dimensions, all zeros except one
   b. 300 dimensions, all zeros
   c. 50,000 dimensions, all ones
   d. 300 dimensions, random values

2. **Main problem with one-hot encoding for large vocabularies:**
   a. Too few dimensions
   b. Vectors are dense
   c. No notion of semantic similarity
   d. Easy to compute similarity

3. **Word embeddings are:**
   a. Sparse binary vectors
   b. Dense low-dimensional vectors
   c. One-hot encoded vectors
   d. Randomly initialized fixed vectors

4. **Distributional hypothesis states:**
   a. Words have fixed meanings
   b. Words are defined by context
   c. All words are equally frequent
   d. Words are independent

5. **Famous word analogy example:**
   a. king - man + woman ≈ queen
   b. king + queen ≈ royalty
   c. man + woman ≈ person
   d. king - queen ≈ man

6. **Word2Vec training uses:**
   a. Co-occurrence matrix only
   b. Sliding window over corpus
   c. Entire corpus matrix
   d. None of the above

7. **Skip-gram predicts:**
   a. Target from context
   b. Context from target
   c. Next word only
   d. Previous word only

8. **CBOW predicts:**
   a. Target from context
   b. Context from target
   c. Next sentence
   d. Document topic

9. **Softmax denominator problem:**
   a. Too small vocabulary
   b. Sum over all $|V|$ words (expensive)
   c. Only positive samples
   d. No gradients

10. **Negative sampling uses:**
    a. 1 positive + K negative samples
    b. Only positive samples
    c. Only negative samples
    d. All vocabulary words

11. **In negative sampling, K is typically:**
    a. 1-2
    b. 5-20
    c. 100-200
    d. 1000+

12. **Negative words are sampled based on:**
    a. Uniform distribution
    b. Word frequency $f(w)^{3/4}$
    c. Alphabetical order
    d. Random order

13. **GloVe uses which matrix?**
    a. Word-context co-occurrence
    b. Term-document
    c. Confusion
    d. Covariance

14. **GloVe loss function is:**
    a. Binary cross-entropy
    b. Weighted MSE
    c. Hinge loss
    d. KL divergence

15. **In GloVe, $f(X_{ij})$ weighting function:**
    a. Downweights rare pairs
    b. Upweights rare pairs
    c. Ignores frequent pairs
    d. Is constant

16. **GloVe takes $\log X_{ij}$ because:**
    a. X_{ij} follows power law
    b. To make values negative
    c. To increase sparsity
    d. To reduce vocabulary size

17. **Word2Vec vs GloVe: Word2Vec uses:**
    a. Global co-occurrence
    b. Local context window
    c. Matrix factorization
    d. No training

18. **Word2Vec vs GloVe: GloVe uses:**
    a. Local context window
    b. Negative sampling
    c. Global co-occurrence matrix
    d. Next word prediction

19. **After Word2Vec training, embeddings are taken from:**
    a. Output layer weights
    b. Input layer weights (W_1)
    c. Bias terms only
    d. Sum of input and output

20. **Distributional hypothesis is credited to:**
    a. Turing
    b. Firth (1957)
    c. Chomsky
    d. Hinton

Answer Key (For Instructor Use Only)

1. A 2. C 3. B 4. B 9. B 10. A 11. B 12. B 17. B 18. C 19. B 20. B
5. A 6. B 7. B 8. A 13. A 14. B 15. A 16. A